

1)

```

> Fertilizer_A=c(55, 60, 50, 57)
> Fertilizer_B=c(62, 68, 63, 60)
> Fertilizer_C=c(53, 53, 52, 54)
> Fertilizer_D=c(68, 70, 120, 5)
> platns=data.frame(Fertilizer_A,Fertilizer_B,Fertilizer_C,Fertilizer_D)
> plants
  FertilizerA FertilizerB FertilizerC FertilizerD
1          55          62          53          68
2          60          68          53          70
3          50          63          52         120
4          57          60          54           5

> tablenew=stack(plants[,c("FertilizerA","FertilizerB","FertilizerC","FertilizerD")])
> colnames(tablenew)=c("heights","Fertilizer")
> tablenew
  heights Fertilizer
1       55 FertilizerA
2       60 FertilizerA
3       50 FertilizerA
4       57 FertilizerA
5       62 FertilizerB
6       68 FertilizerB
7       63 FertilizerB
8       60 FertilizerB
9       53 FertilizerC
10      53 FertilizerC
11      52 FertilizerC
12      54 FertilizerC
13      68 FertilizerD
14      70 FertilizerD
15     120 FertilizerD
16       5 FertilizerD

```

Normality checking

H0: Data are normally distributed

H1: Data are not normally distributed

```

> attach(tablenew)
> shapiro.test(heights)

```

Shapiro-wilk normality test

```

data: heights
W = 0.76704, p-value = 0.001027

```

Since $p\text{ value} < 0.05$, H0 is rejected at 5% level of significance. Therefore the data are not normally distributed. So we have to perform suitable non-parametric test.

Hypothesis

H0: The distributions of plant heights are identical across all fertilizer groups. (There is no significant difference in the mean plant heights among the four fertilizer treatments.)

H1: At least one fertilizer treatment produces systematically different plant heights than the others. (There is a significant difference in the mean plant heights among the four fertilizer treatments.)

```
> kruskal.test(heights~Fertilizer,data=tablenew)
```

Kruskal-wallis rank sum test

data: heights by Fertilizer

Kruskal-wallis chi-squared = 6.5196, df = 3, p-value = 0.08889

Since p value > 0.05, H0 do not reject at 5% level of significance. Therefore there is no evidence that the fertilizer has a significant effect on plant height at the 5% level of significance.

2)

```
> MethodA=c(72, 75, 78, 74, 76)
> MethodB=c(68, 70, 72, 69, 71)
> MethodC=c(80, 82, 85, 83, 81)
> MethodD=c(65, 67, 66, 64, 68)
> riceyield=data.frame(MethodA,MethodB,MethodC,MethodD)
> riceyield
```

	MethodA	MethodB	MethodC	MethodD
1	72	68	80	65
2	75	70	82	67
3	78	72	85	66
4	74	69	83	64
5	76	71	81	68

```
> irrigation=stack(riceyield[,c("MethodA","MethodB","MethodC","MethodD")])
> colnames(irrigation)=c("kg_perplot","irrigation_method")
> irrigation
```

	kg_perplot	irrigation_method
1	72	MethodA
2	75	MethodA
3	78	MethodA
4	74	MethodA
5	76	MethodA
6	68	MethodB
7	70	MethodB
8	72	MethodB
9	69	MethodB
10	71	MethodB
11	80	MethodC
12	82	MethodC
13	85	MethodC

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14	83	MethodC
15	81	MethodC
16	65	MethodD
17	67	MethodD
18	66	MethodD
19	64	MethodD
20	68	MethodD

a)

Normality checking

H0: Data are normally distributed

H1: Data are not normally distributed

```
> shapiro.test(kg_perplot)
```

Shapiro-wilk normality test

data: kg_perplot

w = 0.94742, p-value = 0.3296

Since p value > 0.05, H0 is not rejected at 5% level of significance. Therefore the data are normally distributed. So we have to perform ANOVA test.

Hypothesis

H0: irrigation method has no significant effect on rice crop yield.

H1: irrigation method has a significant effect on rice crop yield.

```
> attach(irrigation)
```

```
> model1=aov(kg_perplot~irrigation_method,data=irrigation)
```

```
> summary(model1)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
irrigation_method	3	731.4	243.80	71.18	1.8e-09 ***
Residuals	16	54.8	3.43		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Since P value < 0.05, H0 is rejected at 5% level of significance. Therefore we can conclude that irrigation method has an effect on crop yield at the 5% level of significance.

b)

Hypothesis

 $H_0: \mu_i = \mu_j$

$H_1: \mu_i \neq \mu_j$; where μ_i = mean rice kg per plot when using different irrigation techniques and $i \& j \in \{A, B, C, D\}$

```
> TukeyHSD(model1)
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = kg_perplot ~ irrigation_method, data = irrigation)
```

```
$irrigation_method
      diff      lwr      upr    p adj
MethodB-MethodA -5.0 -8.348738 -1.6512622 0.0029468
MethodC-MethodA  7.2  3.851262 10.5487378 0.0000748
MethodD-MethodA -9.0 -12.348738 -5.6512622 0.0000051
MethodC-MethodB 12.2  8.851262 15.5487378 0.0000001
MethodD-MethodB -4.0 -7.348738 -0.6512622 0.0167017
MethodD-MethodC -16.2 -19.548738 -12.8512622 0.0000000
```

Since the all the p values less than 0.05, we can reject H_0 at 5% level of significance. Therefore we can conclude that mean rice kg per plot when using each irrigation techniques are significantly different.

c)

Normality checking

H_0 : residuals are normally distributed.

H_1 : residuals are not normally distributed.

```
> gg=residuals(model1)
> shapiro.test(gg)
```

Shapiro-wilk normality test

data: gg

w = 0.96784, p-value = 0.7087

Since P value > 0.05, we do not reject H_0 at 5% level of significance. Therefore residuals are normally distributed .

Homogeneity checking

H_0 : residuals have constant variance

H_1 : residuals haven't constant variance

```
> leveneTest(model1)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  3   0.163 0.9197
      16
```

Since P value > 0.05, we do not reject H0 at 5% level of significance. Therefore residuals have constant variance.

3)

```
> drug=read.table(file.choose(),header=T)
> newdrug=stack(drug[,c("Drug_A","Drug_B","Drug_C","Drug_D")])
> colnames(newdrug)=c("blood_pressure","drug_type")
```

```
> newdrug
      blood_pressure drug_type
1              12   Drug_A
2              15   Drug_A
3              14   Drug_A
4              13   Drug_A
5              16   Drug_A
6              14   Drug_A
7              18   Drug_B
8              20   Drug_B
9              19   Drug_B
10             17   Drug_B
11             21   Drug_B
12             18   Drug_B
13             10   Drug_C
14             11   Drug_C
15              9   Drug_C
16             12   Drug_C
17             10   Drug_C
18             11   Drug_C
19             14   Drug_D
20             16   Drug_D
21             15   Drug_D
22             13   Drug_D
23             17   Drug_D
24             16   Drug_D
```

A.

Normality checking

H0: Data are normally distributed

H1: Data are not normally distributed

```
> shapiro.test(blood_pressure)
```

Shapiro-wilk normality test

```
data: blood_pressure
w = 0.97702, p-value = 0.8352
```

Since $p\text{ value} > 0.05$, H_0 is not rejected at 5% level of significance. Therefore the data are normally distributed. So we have to perform ANOVA test.

Hypothesis

H_0 : There is no significant difference in effectiveness among the drugs.

H_1 : There is a significant difference in effectiveness among the drugs.

```
> model2=aov(blood_pressure~drug_type,data=newdrug)
> summary(model2)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
drug_type	3	212.46	70.82	38.11	1.84e-08 ***
Residuals	20	37.17	1.86		

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since $P\text{ value} < 0.05$, H_0 is rejected at 5% level of significance. Therefore we can conclude that there is a significant difference in effectiveness among the drugs.

B.

Hypothesis

$H_0: \mu_i = \mu_j$

$H_1: \mu_i \neq \mu_j$;

Where μ_i = mean blood pressure when using different drug type
and $i \& j \in \{A, B, C, D\}$

```
> TukeyHSD(model2)
```

```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = blood_pressure ~ drug_type, data = newdrug)
```

\$drug_type		diff	lwr	upr	p adj
Drug_B-Drug_A	4.833333	2.630436	7.036230	0.0000295	
Drug_C-Drug_A	-3.500000	-5.702897	-1.297103	0.0013050	
Drug_D-Drug_A	1.166667	-1.036230	3.369564	0.4660167	
Drug_C-Drug_B	-8.333333	-10.536230	-6.130436	0.0000000	
Drug_D-Drug_B	-3.666667	-5.869564	-1.463770	0.0008050	

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Drug_D-Drug_C 4.666667 2.463770 6.869564 0.0000468

From the output ,

Since only the P value of Drug_D-Drug_A is greater than 0.05 , therefore H0 is not rejected for that pair.

Therefore the mean blood pressure using drug A equal to using drug D.

Since the all other the p values less than 0.05, we can reject H0 at 5% level of significance for those pair

Therefore we can conclude that mean blood pressure when using each drug type are significantly different.

C.

Normality checking

H0: residuals are normally distributed.

H1: residuals are not normally distributed.

```
> yy=residuals(model2)
```

```
> shapiro.test(yy)
```

Shapiro-wilk normality test

data: yy

w = 0.97109, p-value = 0.6939

Since P value>0.05, we do not reject H0 at 5% level of significance. Therefore residuals are normally distributed .

Homogeneity checking

H0: residuals have constant variance

H1: residuals haven't constant variance

```
> leveneTest(model2)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	0.2546	0.8571
	20		

Since P value>0.05, we do not reject H0 at 5% level of significance. Therefore residuals have constant variance.

4)

```

> treatment_1=c( 55, 60, 58, 57);
> treatment_2=c( 62, 65, 63, 66);
> treatment_3=c( 51, 53, 52, 54);
> treatment_4=c( 68, 95, 40, 120);
> Treat=data.frame(treatment_1,treatment_2,treatment_3,treatment_4)

> Treatment=stack(Treat[,c("treatment_1","treatment_2","treatment_3","treatment_4")])
> colnames(Treatment)=c("data","treatment_type")
> Treatment
  data treatment_type
1    55    treatment_1
2    60    treatment_1
3    58    treatment_1
4    57    treatment_1
5    62    treatment_2
6    65    treatment_2
7    63    treatment_2
8    66    treatment_2
9    51    treatment_3
10   53    treatment_3
11   52    treatment_3
12   54    treatment_3
13   68    treatment_4
14   95    treatment_4
15   40    treatment_4
16  120    treatment_4

```

I.

Hypothesis

H0: There is no significant difference in the mean responses among the four treatment groups

H1: There is a significant difference in the mean responses among the four treatment groups

```

> model3=aov(data~treatment_type,data=Treatment)
> summary(model3)
              Df Sum Sq Mean Sq F value Pr(>F)
treatment_type  3   1819    606.2    2.024  0.164
Residuals      12   3595    299.6

```


Since p value > 0.05 , therefore H_0 is not rejected at 5% level of significance. Therefore we can conclude that There is no significant difference in the mean responses among the four treatment groups.

II.

Normality checking

H_0 : residuals are normally distributed.

H_1 : residuals are not normally distributed.

```
> xx=residuals(model3)
> shapiro.test(xx)
```

Shapiro-wilk normality test

```
data: xx
W = 0.75481, p-value = 0.0007263
```

Since P value < 0.05 , we reject H_0 at 5% level of significance. Therefore residuals are not normally distributed .

Homogeneity checking

H_0 : residuals have constant variance

H_1 : residuals haven't constant variance

```
> leveneTest(model3)
Levene's Test for Homogeneity of Variance (center = median)
  Df F value    Pr(>F)
group 3   10.88 0.0009706 ***
    12
```

Since P value < 0.05 , we reject H_0 at 5% level of significance. Therefore residuals have not constant variance.

III.

An alternative method for violating homogeneity

```
> oneway.test(data~ treatment_type, data =Treatment)
```

One-way analysis of means (not assuming equal variances)

```
data: data and treatment_type
F = 29.896, num df = 3.0000, denom df = 6.2683, p-value =
0.0004196
```

Again, Since P value < 0.05 , we reject H_0 at 5% level of significance. Therefore residuals have not constant variance.